Thermal radiation of Leslie's cube and a black body

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The T^4 dependency of the Stefan-Boltzmann law has been verified via measurements on the Leslie's cube and on a cavity which simulated a blackbody. The infrared transmittance of window glass, silicon and sodium chloride have been measured to $t_{glass} = (2.27 \pm 0.48)\%$, $t_{silicon} = (46.82 \pm 0.48)\%$ and $t_{sodiumchloride} = (73.32 \pm 0.48)\%$ respectively. Further the angle dependency of the Lambert's law for a cavity radiator at $T = 371°\mathrm{C}$ has been tested and validated.

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1 Introduction

Every object with a temperature above absolut zero emits electromagnetic waves in the form of thermal radiation. In an thermodynamical equilibrium this radiation is described by the Stefan-Boltzmann law.
Generally it is important in many field how different materials absorb or emit radiation - for example in construction it is necessary for insulation so that as little thermal radiation is absorbed. The respective abilities of the materials used depend heavily on their properties and internal structure - for effective application, therefore, knowledge of the material-specific properties is important.
In the following those characteristics for different surfaces will be examined.

2 Theory

In the following sections we will examine the emission of different surfaces and of a cavity radiator. A reasonable theoretical starting point is the black body radiator. Its spectral density of electromagnetic radiation is given by Planck's law:

$$I(\nu, T) = \frac{2h\nu^3}{c^2} \cdot \frac{1}{e^{\frac{h\nu}{kT}} - 1} \tag{1}$$

Here the T is the temerature of the black body, ν is the frequency of the emitted radiation, h is Planck's constant, c is the speed of light and k is Boltzmann's constant. The total power emitted by a black body per unit area can be calculated by integrating over all frequencies and the solid angle:

$$\frac{P}{A} = \int_0^\infty I(\nu, T) d\nu \int cos(\theta) d\Omega \tag{2}$$

The cosine is there because the black body is also a Lambertian emitter, meaning that the intensity of the emitted radiation is proportional to the cosine of the angle between the normal of the surface and the direction of emission. Computing the integral yields the Stefan-Boltzmann law:

$$P = \sigma A T^4 \tag{3}$$

Where σ is the Stefan-Boltzmann constant:

$$\sigma = \frac{2\pi^5 k^4}{15 h^3 c^2} \approx 5.67 \cdot 10^{-8} \frac{W}{m^2 K^4} \tag{4}$$

Since the black body is an idealized object, Eq. 3 is not valid for real objects. Instead, we have to introduce the specific emissivity ϵ of the object, which is smaller than 1 for real objects and equals 1 for a perfect black body. The modified Stefan-Boltzmann law for real objects is then:

$$P = \epsilon \sigma A T^4 \tag{5}$$

For the following measurments it is important to note, that the Leslie's cube also absorbs radiation from the environment and re-emits it with respect to the Kirchhoff's law. Kirchhoff's law states that the emissivity ϵ of a surface is equal to its absorptivity α. Therefore the radiation generated only by the temperature of the cube is the difference between the emitted radiation and the absorbed radiation. T_0 is the temperature of the environment, which we will assume to be room temperature. It follows:

$$P \propto T^4 - T_0^4 \tag{6}$$

3 Experiment

3.1 Measurements on Leslie's cube

For the following measurements the setup shown in figure 1 was used. This consists of a Leslie's cube on a turntable which is heated by a thermostat. The cube has four different surfaces: a white, a black, a shiny nickel-plated and a matte nickel-plated. A thermopile is used to measure the thermal radiation emitted by the different surfaces of the cube. The thermopile is connected to a voltmeter which displays the voltage generated by the thermopile. The filtermount can be used to place different filters in front of the thermopile, which will be done in section 3.1.2. The errors of the temperature and the voltage are calculated according to the specifications of the thermostat Lauda Kältethermostate RE 104 and the voltmeter Agilent 34405A ([1], [2]).

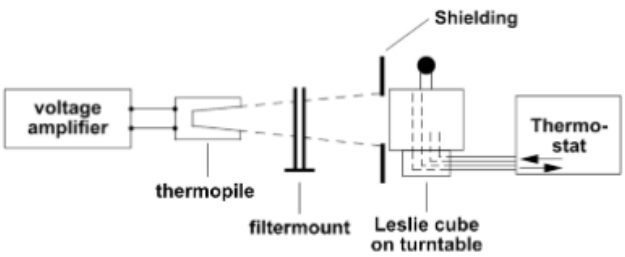

Figure 1: Experimental setup for measurements on Leslie's cube.

3.1.1 Stefan-Boltzmann-law

Figure 2 shows the voltage generated by the thermopile as a function of the temperature of the cube for the different surfaces. The fit used is of the form $U = A(T^b - T_0^b)$, where $T_0 = 293.15K$ is the reference temperature measured via the zero point of the voltmeter. The fit parameters are shown in table **??**. As one can see in figure 2 the black side emits the most thermal radiation and the polished nickel-plated side emits the least, which is expected since the black surface has the highest emissivity and the polished nickel-plated surface has the lowest emissivity. The data of the polished nickel-plated surface is very noisy. This matches with it's fit exponent b which is far off aswell, this will be discussed further below in more detail.

Surface	fit exponent b
White	3.770 ± 0.059
Black	4.043 ± 0.081
Polished nickel-plated	2.8 ± 2.3
Matte nickel-plated	3.94 ± 0.21

Table 1: The exponents determined by the $U = A(T^b - T_0^b)$ fit to the data obtained by measurements on Leslie's cube.

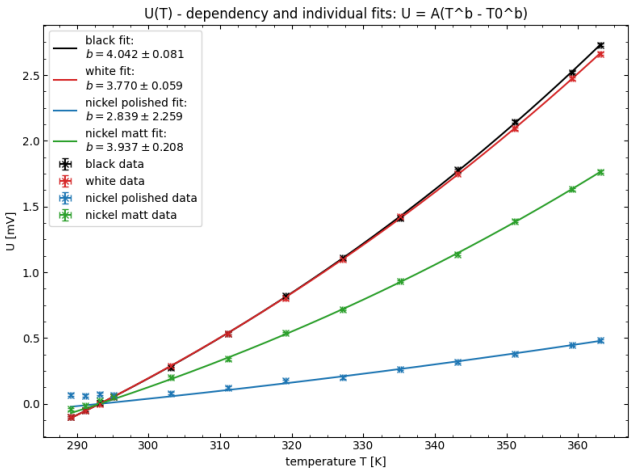

Figure 2: Voltage generated by the thermopile as a function of the temperature of the cube for the different surfaces. The fit used is of the form $U = A(T^b - T_0^b)$. With $T_0 = 293.15K$ as the reference temperature.

The expected values of the exponents is $b = 4$ according to the Stefan-Boltzmann law. The values obtained for the black and matte nickel-plated surfaces match that within their error, while the value obtained for the white and polished nickel-plated surface are a bit off. The value obtained for the white surface is close to the expected value, but the error is small enough to exclude $b = 4$. The value obtained for the polished nickel-plated does match $b = 4$ within the error, but the error is very large which is a good representation of the poor quality of the measurement. This poor quality is due to the fact that the polished side reflects a lot of thermal radiation from the environment, which is also measured by the thermopile and can not be easily seperated from the radiation emitted by the surface itself. A solution to this would be to place the setup in a more contained environment and shield it further from external radiation.

The voltage is also plotted as a function of the forth power of the temperature as you can see in figure 3. Here one can see that all sides follow a linear trend, which is expected. The slope of the nickel-plated surfaces is much smaller than the slope of the white and black surfaces, which is also expected since they have a much lower emissivity.

3.1.2 Infrared permeability for several materials

In this part of the experiment the setup in figure 1 was reused, but only the black side of the cube was used, now with different optical filters placed in front of the thermopile. The used filters were: window glass, silicon and sodium chloride.
The temperature was set to $T = 77°\mathrm{C}$ with an en-

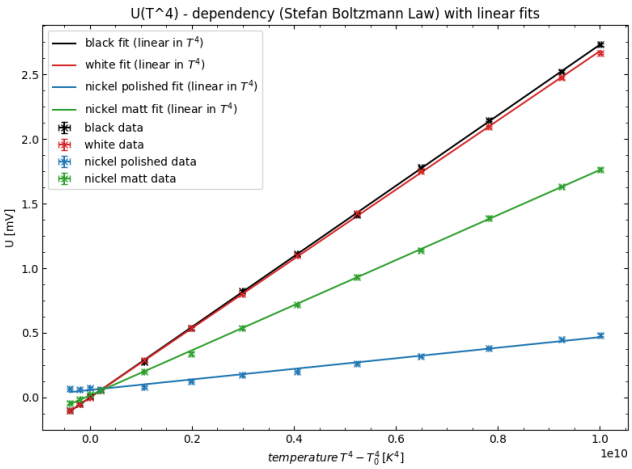

Figure 3: Voltage generated by the thermopile as a function of the forth power of the temperature of the cube for the different surfaces. The fit used is of the form $U = A(T^4 - T_0^4)$, where T_0 is the room temperature.

Filter	infrared permeability (%)
Window glass	2.27 ± 0.48
Silicon	46.82 ± 0.48
Sodium chloride	73.32 ± 0.48

Table 2: Transmittance of infrared radiation through different filters at $T = 77°\mathrm{C}$.

vironmental temperature of $T_0 = 20°\mathrm{C}$. The thermal radiation emitted by the black surface was measured first, after that the filters were placed in front of the thermopile and the measurements were repeated. The results are given in percentages of the voltage measured without a filter and shown in table 2.
It shows that thus although the window glass transmits visible light very well, the infrared transmittance is the lowest of the three. Since glass is an amorphous material, it has a lot of vibrational modes which can be excited by the infrared radiation, which leads to a high absorption and thus low transmittance. Silicone however lets almost half of the infrared radiation through, which is due to the fact that it is a crystalline material with covalent bonds and to be more precise, it is a indirect band gap semiconductor. This means that it has a band gap which can be excited by the infrared radiation, but for cristal-momentum conservation, a phonon also needs to be involved, which is less likely to happen than just exciting the band gap, which leads to a higher transmittance than glass but not as high as sodium chloride. Sodium chloride has the highest transmittance of the three, which can be referred to the fact that it is an ionic crystal with a large band gap, which can not be excited by the infrared radiation, leading to a very low absorption and thus high transmittance.
With Wien's displacement law one can also calculate

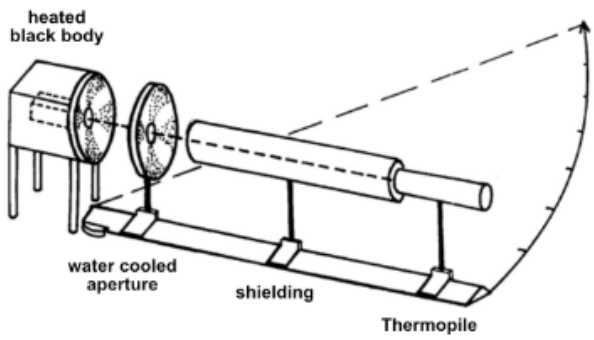

Figure 4: Experimental setup for measurements on blackbody radiation.

the wavelength of the maximum spectral intensity of the thermal radiation:

$$\lambda_{max} = \frac{b}{T} \tag{7}$$

where $b = 2897.771955 \mu m * K$ is Wien's displacement constant and T is the temperature of the black surface in Kelvin. For the measured room temperature of $T_0 = 20°\mathrm{C}$ and the temperature of the black surface of $T = 77°\mathrm{C}$, one obtains a wavelength of $\lambda_{max,0} = (9.885 \pm 0.033 \mu m)$ and $\lambda_{max,leslie} = (8.276 \pm 0.020 \mu m)$ respectively. Both are in the infrared range, which is consistent with the fact that the filters used in this part of the experiment are designed to filter infrared radiation.

3.2 Blackbody radiation

The setup of the following part is seen in figure 4. It of an electric oven in which a hollow cylinder is heated. Through a hole in the cavity the thermal radiation is emitted. This is a close approximation of a blackbody, since a black body isn't possible in reality.

3.2.1 Stefan-Boltzmann-law

Analogously to 3.1, the voltage generated by the thermopile is measured in an temperature range of Room temperature $= 20°\mathrm{C} \leq T \leq 400°\mathrm{C}$. The angle at which the thermopile is placed is chosen to be $\phi = 0°$, so it is directly facing the hole of the hollow cylinder. The data is again first fitted with the function $U = A(T^b - T_0^b)$, where T_0 is the room temperature. The fit is shown in figure 5 and the fit parameter turns out to be $b = 4.68 \pm 0.16$.
The obtained value of b is quite far off from the expected value of $b = 4$ according to the Stefan-Boltzmann law. This could be due to the fact that the thermopile is not only measuring the thermal radiation emitted by the blackbody, but also the thermal

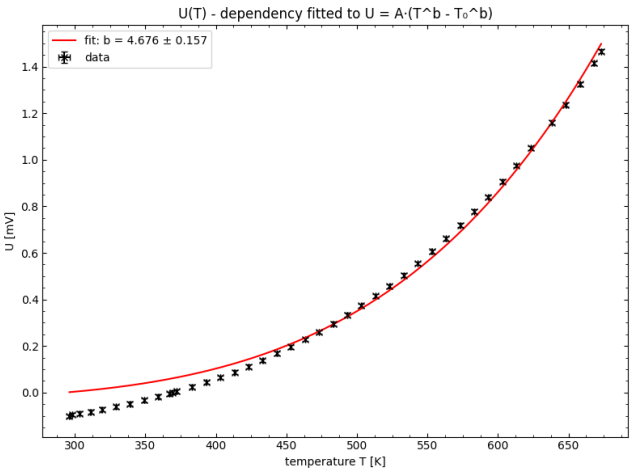

Figure 5: Voltage generated by the thermopile as a function of the temperature of the blackbody for $\phi = 0°$. The fit used is of the form $U = A(T^b - T_0^b)$, where T_0 is the room temperature.

radiation emitted by the environment and reflected on the walls of the cavity, which is not negligible at room temperature. Nevertheless it is still in the same order of magnitude as the expected value, which is a good indication that the Stefan-Boltzmann law is still valid for this setup.
The voltage is again also plotted as a function of the forth power of the temperature as you can see in figure 6.

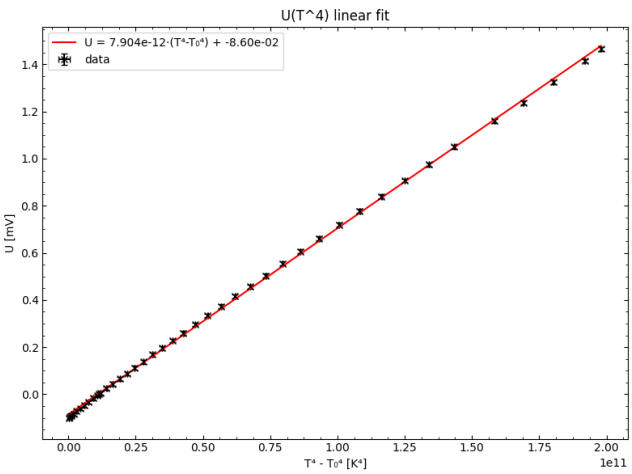

Figure 6: Voltage generated by the thermopile as a function of the forth power of the temperature of the blackbody for $\phi = 0°$. The fit used is of the form $U = A(T^4 - T_0^4)$, where T_0 is the room temperature.

Again here one can see that the data follows a linear trend, which indicates that the Stefan-Boltzmann law is valid for this setup, even though the obtained value of b is quite a bit off from the expected value.

3.2.2 Lambert's cosine law

Since blackbodies radiate without any preferred direction, the intensity of the radiation emitted by a blackbody should follow Lambert's cosine law:

$$I(\phi) = I_0 \cos(\phi) \tag{8}$$

The angle ϕ is defined as the angle between the normal vector of the surface and the direction in which the intensity is measured.

The setup for this part is the same as in 3.2.1 but now the angle ϕ at which the thermopile is placed is varied in the range of $0° \leq \phi \leq 50°$ and the temperature is kept constant at $T = 371°$C. The measured voltage is first normalized to the voltage measured at $\phi = 0°$ and then plotted as a function of the angle ϕ and fitted with the trigonometric function $U = a\cos(b\phi + c) + d$. The ideal fit is also included in the plot following the function $U = a\cos(\phi) + b$. The fit is shown in figure 7. [H] Then measured voltage

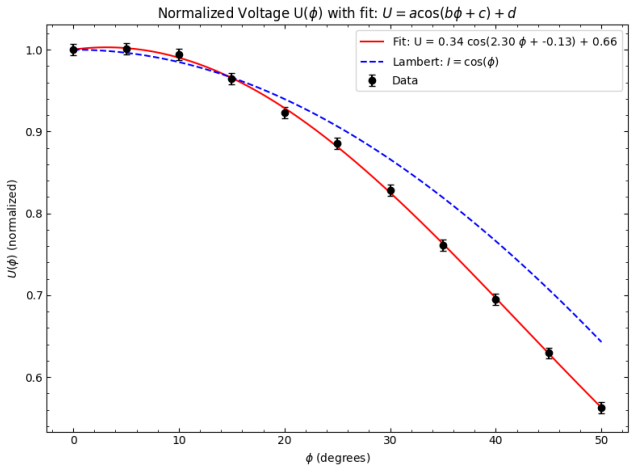

Figure 7: Voltage generated by the thermopile as a function of the angle ϕ for $T = 371°$C. The voltage is normalized to the voltage measured at $\phi = 0°$.

is also plotted as a function of $\cos(\phi)$ and fitted with the function $U = a\cos(\phi) + b$, where a and b are the fit parameters. Again the ideal fit is also included in the plot following the function $U = a\cos(\phi) + b$. The fit is shown in figure 8.
In figure 7 one can see that the data follows a cosine trend at small angles, but at larger angles it deviates from the cosine trend. This could be due to the fact that at larger angles, the thermopile is not only measuring the radiation emitted by the blackbody, but also the radiation emitted by the environment and reflected on the walls of the cavity, which is not negligible at room temperature.
In figure 8 one can see that the data follows a linear trend and has a similar slope as the ideal fit, which is a good indication that the Lambert's cosine law

is valid for this setup, even though there are some deviations at larger angles.

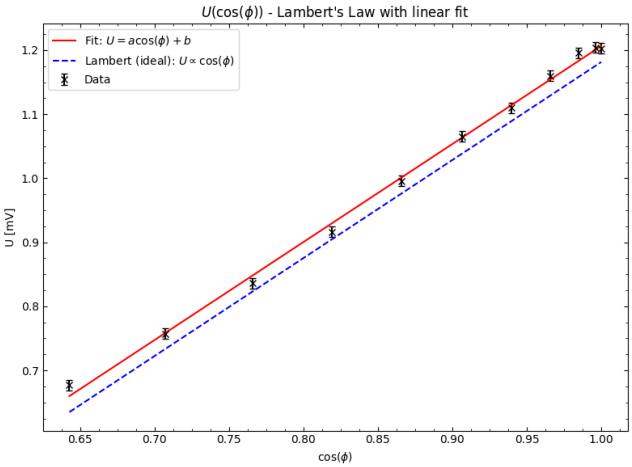

Figure 8: Voltage generated by the thermopile as a function of $\cos(\phi)$ for $T = 371°\mathrm{C}$. The fit used is of the form $U = a\cos(\phi) + b$, where a and b are the fit parameters.

4 Conclusion

The Stefan-Boltzmann law has been validated for two different sources of thermal radiation, namly the Leslie's cube and a simulated blackbody.
The fit exponents of the different surfaces of the Leslie's cube turned out to be $b_{white} = 3.770 \pm 0.059$ for the white side, $b_{black} = 4.043 \pm 0.081$ for the black side, $b_{polished} = 2.8 \pm 2.3$ for the polished nickel-plated side and $b_{matte} = 3.94 \pm 0.21$ for the matte nickel-plated side. Exept the polished side the values for the fit exponents were plausible. The problem with the polished side was the missing shielding from environmental radiation. The fit exponent of the blackbody resulted in $b_{blackbody} = 4.68 \pm 0.16$ with is also a bit off, but is still in the same magnitude. These fit exponents suggest, that the Stefan-Boltzmann law is valid for both the Leslie's cube and the blackbody.

For different materials placed in the pathway between the thermopile and the source of thermal radiation, the transmittance can be calculated relative to a free path. These resulted in $t_{glass} = (2.27 \pm 0.48)\%$ for window glass, $t_{silicon} = (46.82 \pm 0.48)\%$ for silicon and $t_{sodiumchloride} = (73.32 \pm 0.48)\%$ which all match with the theoretical considerations.
According to Wien's law the maximal thermal radiation wavelength can be calculated. $\lambda_{max,0} = (9.885 \pm 0.033)\mu m$ for the room temperature and $\lambda_{max,leslie} = (8.276 \pm 0.020\mu m)$ for the Leslie's cube. If the blackbody is not observed head-on, but at an angle, the intensity is expected to follow Lambert's cosine law. The thermal radiation emitted by the oven was found to approximately follow this law especially for small angles.

References

[1] `https://www.physik.uni-wuerzburg.de/fileadmin/11000000/03_Studium/02_Bachelor/Grundpraktikum/_imported/fileadmin/11016800/Geraete/RE104.pdf` last visited on 24.04.2026

[2] `https://www.physik.uni-wuerzburg.de/fileadmin/11000000/03_Studium/02_Bachelor/Grundpraktikum/_imported/fileadmin/11016800/Geraete/Agilent-34405a.pdf` last visited on 24.04.2025